DIG - Domain Information Groper

It is a command-line tool used to query **DNS (Domain Name System) Name Servers**. It is the industry standard for troubleshooting because it shows you the "raw" data coming back from the servers, unlike simpler tools like `nslookup`.

What does it actually do?

At Level Zero, `dig` performs four main tasks for you:

1. The Translation (Name to IP)

Its primary job is telling you where a domain lives.

- **Input:** `dig yahoo.com`
- **Output:** `74.6.231.21` (The IPv4 address).

2. Identifying "Ownership" (Name servers)

It tells you which specific server is the "boss" of a domain.

- If you want to know who manages the records for your lab, you ask for the **NS (Name Server)** records.

3. Checking "Freshness" (TTL)

Every piece of DNS info has an expiration date called **TTL (Time to Live)**. `dig` shows you exactly how many seconds are left before a server is required to throw away that info and ask for a fresh copy. This is why when you change an IP, it doesn't happen instantly for everyone.

4. Path Tracing

Using the `+trace` flag (which you had in your notes), `dig` acts like a GPS. It starts at the very top of the internet (the **Root Servers**) and follows the path down through the country code (like `.bd`) until it hits the specific server in your lab.

```
fedora@fedora:~$ dig yahoo.com

 <<>> DiG 9.18.44 <<>> yahoo.com
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 40381
; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

; OPT PSEUDOSECTION:
 EDNS: version: 0, flags:; udp: 65494
; QUESTION SECTION:
yahoo.com.                      IN      A

; ANSWER SECTION:
yahoo.com.              1363    IN      A       74.6.231.21
yahoo.com.              1363    IN      A       74.6.231.20
yahoo.com.              1363    IN      A       74.6.143.25
yahoo.com.              1363    IN      A       98.137.11.163
yahoo.com.              1363    IN      A       74.6.143.26
yahoo.com.              1363    IN      A       98.137.11.164

; Query time: 157 msec
; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
; WHEN: Wed Feb 18 19:28:57 +06 2026
; MSG SIZE  rcvd: 134
```

Here is a concise, line-by-line breakdown of your `dig` output:

The Request Metadata

- `DiG 9.18.44`: The version of the `dig` tool you are using.
- `global options: +cmd`: Confirms that the command-line arguments are being processed.
- `Got answer:`: Confirms the DNS server responded successfully.

The Header Section

- `opcode: QUERY`: You performed a standard "Question" lookup.
- `status: NOERROR`: Success; the domain name exists and was found.
- `id: 40381`: A random number used to match this specific answer to your request.
- `flags: qr rd ra`:
  - `qr` **(Query Response)**: This is a reply from a server.
  - `rd` **(Recursion Desired)**: You asked the server to do the hard work of finding the IP.
  - `ra` **(Recursion Available)**: The server confirmed it is capable of doing that work.
- `QUERY: 1, ANSWER: 6, ...`: You asked 1 question and received 6 different IP addresses in response.

The Question Section

- `;yahoo.com. IN A`: This restates your question: "What is the **A** (IPv4) record for **yahoo.com**?"

The Answer Section

- `yahoo.com.`: The name being looked up.
- `1363`: **TTL (Time to Live)**. The number of seconds remaining before this info expires from your local cache.
- `IN`: "Internet" (the class of the network).
- `A`: The record type (**Address**).
- `74.6.231.21`: The physical IPv4 address of the Yahoo server.

The Footer (Stats)

- `Query time: 157 msec`: How long the round-trip request took.
- `SERVER: 127.0.0.53#53`: The IP and port of the machine that answered you (your local system resolver).
- `WHEN: Wed Feb 18...`: The exact timestamp of the query.
- `MSG SIZE rcvd: 134`: The size of the DNS response packet in bytes.

Name server Lookup

```
dig yahoo.com NS
```

```
fedora@fedora:~$ dig yahoo.com NS

; <<>> DiG 9.18.44 <<>> yahoo.com NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 52020
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;yahoo.com.                     IN      NS

;; ANSWER SECTION:
yahoo.com.              21058   IN      NS      ns2.yahoo.com.
yahoo.com.              21058   IN      NS      ns3.yahoo.com.
yahoo.com.              21058   IN      NS      ns4.yahoo.com.
yahoo.com.              21058   IN      NS      ns1.yahoo.com.
yahoo.com.              21058   IN      NS      ns5.yahoo.com.

;; Query time: 133 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Feb 18 19:40:07 +06 2026
;; MSG SIZE  rcvd: 128
```

Directed Query

```
fedora@fedora:~$ dig @ns1.yahoo.com yahoo.com

; <<>> DiG 9.18.44 <<>> @ns1.yahoo.com yahoo.com
; (2 servers found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 55643
;; flags: qr aa rd; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1272
; COOKIE: 8b579a3ea3a2786b010000006995c25ce89801cdb66e4b2b (good)
;; QUESTION SECTION:
;yahoo.com.                     IN      A

;; ANSWER SECTION:
yahoo.com.              1800    IN      A       74.6.231.20
yahoo.com.              1800    IN      A       74.6.231.21
yahoo.com.              1800    IN      A       74.6.143.25
yahoo.com.              1800    IN      A       98.137.11.164
yahoo.com.              1800    IN      A       98.137.11.163
yahoo.com.              1800    IN      A       74.6.143.26

;; Query time: 206 msec
;; SERVER: 68.180.131.16#53(ns1.yahoo.com) (UDP)
;; WHEN: Wed Feb 18 19:45:00 +06 2026
;; MSG SIZE  rcvd: 162
```

By using `@ns1.yahoo.com`, you told `dig` to ignore your local Fedora settings and send the packet directly across the internet to Yahoo's physical hardware. This is why the **SERVER** line at the bottom shows an external IP (`68.180.131.16`) instead of your usual `127.0.0.53`.

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2. The "aa" Flag (Authoritative Answer)

This is the most important part of your screenshot for your lab.

- Look at the line: `;; flags: qr aa rd;`
- The presence of `aa` means you are talking to the "Source of Truth."
- **The technicality:** The server checked its own local configuration file (its **Zone File**) and found the answer there. It didn't have to ask anyone else or look in its history.

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3. The Recursion Warning

`WARNING: recursion requested but not available`

- This is a "security" feature. Yahoo's nameservers are **Authoritative**, not **Recursive**.
- **Explanation:** They are designed to answer questions *about* Yahoo. If you tried to ask this specific server for the IP of `google.com`, it would refuse. It's telling you: "I can give you my own records, but don't ask me to go find other people's records for you."

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4. The Response Consistency

Notice the **ANSWER SECTION** still shows 6 different IP addresses.

- These are the same IPs you saw in Step 1.
- However, because you asked the "Boss," you are now 100% certain these are the correct, official IPs intended by Yahoo's engineers, with no risk of "cache poisoning" or outdated info from a middleman.

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5. Why this is the "Secret Sauce" for your Lab

In your notes, you have the command: `dig @127.0.0.1(my loopback) www.yahoo.com`.

- If you run that, you are asking **your own Bind9 server** to go find Yahoo.
- In that case, your server acts as a **Recursive Resolver** (the middleman).
- But when you eventually run `dig @127.0.0.1 s16.zhjlab.bd`, you are asking your server about **itself**. You will be looking specifically for that `aa` **flag** to prove you configured your `named.conf.local` correctly.

Dig Trace

```
fedora@fedora:~$ dig +trace yahoo.com

; <<>> DiG 9.18.44 <<>> +trace yahoo.com
;; global options: +cmd
.                       87202   IN      NS      j.root-servers.net.
.                       87202   IN      NS      m.root-servers.net.
.                       87202   IN      NS      g.root-servers.net.
.                       87202   IN      NS      h.root-servers.net.
.                       87202   IN      NS      e.root-servers.net.
.                       87202   IN      NS      k.root-servers.net.
.                       87202   IN      NS      l.root-servers.net.
.                       87202   IN      NS      d.root-servers.net.
.                       87202   IN      NS      c.root-servers.net.
.                       87202   IN      NS      i.root-servers.net.
.                       87202   IN      NS      a.root-servers.net.
.                       87202   IN      NS      f.root-servers.net.
.                       87202   IN      NS      b.root-servers.net.
.                       87202   IN      RRSIG   NS 8 0 518400 20260303050000 20260218040000 21831 . kw/Nwc1eSVGQ5ErVJNtqWuUD0B3qMen+jROemsrSyvJGK6Gqvihq4aRj PL2lY4+lGeXQ1D5vAkH+3h72NAKifTW
ZFUAjBsBb2WpcoEOzAU6GtssV NK9TuXlsAJrD/y3Nk9AFTlq46kx+FtM1XRl5mHAdq/vTNmfvaBDS9EBM C+Z8UdAtEuN393AAsctDz7t0fIE+M0AQgVJN6GIHrKdjqr1wjsryLeQc oNUV7f8F25sXOIZfhInmj+96kfwjill3si/upUei/Nwgm0ca
frpIqWB7 96l0+3KkWP5Yc/iX1yGsGskME6xqhz6F2wmFKVG8Y7Vj02XPL1lL6jyK 7owCUQ==
;; Received 525 bytes from 127.0.0.53#53(127.0.0.53) in 90 ms

com.                    172800  IN      NS      k.gtld-servers.net.
com.                    172800  IN      NS      d.gtld-servers.net.
com.                    172800  IN      NS      j.gtld-servers.net.
com.                    172800  IN      NS      f.gtld-servers.net.
com.                    172800  IN      NS      g.gtld-servers.net.
com.                    172800  IN      NS      l.gtld-servers.net.
com.                    172800  IN      NS      e.gtld-servers.net.
com.                    172800  IN      NS      c.gtld-servers.net.
com.                    172800  IN      NS      a.gtld-servers.net.
com.                    172800  IN      NS      b.gtld-servers.net.
com.                    172800  IN      NS      m.gtld-servers.net.
com.                    172800  IN      NS      h.gtld-servers.net.
com.                    172800  IN      NS      i.gtld-servers.net.
com.                    86400   IN      DS      19718 13 2 8ACBB0CD28F41250A80A491389424D341522D946B0DA0C0291F2D3D7 71D7805A
com.                    86400   IN      RRSIG   DS 8 1 86400 20260303050000 20260218040000 21831 . g+YEw1XKrrUXcDv6u2MBNpfXlU+GVFIVcRyl42tbXlLLx5871nXG8NU9 P387mD+Jj5o7/JwGK3+elryROeFViq+S
9PoX2rOF4chICIJ6nk3Kw2lH C3cFitgXqyvDGE4MiDtQmmNR79UGxl0vfl+BsOsbCe0iBSR1SHP64oTn 0OrgO/k3EtuqQG1HV9SnF31xhRO24qjXAsP8uirMk6oAiH/Rb6FY/Sv5 fTTIyzipfR4hVRVB4UbFb3DfO0Omn1NZgqX7RSK8QEgQaL00V
IuIAakl uUsbt+S+EhW+iyvJPou5TGz8YCVTDPOsSTQcQwXRdRwxZwynxtg/xweH VftrOQ==
;; Received 1197 bytes from 170.247.170.2#53(b.root-servers.net) in 101 ms
```

![[Pasted image 20260218200430.png

```
┌─────────────────────────────────────────────────────────────────────┐
│  STEP 1: ROOT SERVERS (.)                                           │
│  ────────────────────────                                           │
│  Query: "Who handles .com?"                                         │
│  Answer: "Ask a.gtld-servers.net, b.gtld-servers.net, ..."          │
│  Source: Your resolver → Root servers (a.root-servers.net)          │
│  IPs: 198.41.0.4, 192.228.79.201, etc.                              │
└─────────────────────────────────────────────────────────────────────┘

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┌─────────────────────────────────────────────────────────────────────┐
│  STEP 2: TLD SERVERS (.com)                                         │
│  ───────────────────────────                                        │
│  Query: "Who handles yahoo.com?"                                    │
│  Answer: "Ask ns1.yahoo.com, ns2.yahoo.com, ..."                    │
│  Source: Root server → TLD servers (a.gtld-servers.net)             │
│  IPs: 192.5.6.30, 192.33.14.30, etc.                                │
└─────────────────────────────────────────────────────────────────────┘

                                    ↓

┌─────────────────────────────────────────────────────────────────────┐
│  STEP 3: AUTHORITATIVE SERVERS (yahoo.com)                          │
│  ──────────────────────────────────────────                         │
│  Query: "What's the A record for yahoo.com?"                        │
│  Answer: "98.137.11.163, 74.6.231.20, ..."                          │
│  Source: TLD server → Yahoo's NS (ns1.yahoo.com)                    │
│  IP: 68.180.131.16                                                  │
└─────────────────────────────────────────────────────────────────────┘
```

The process

```
┌──────────────────────────────────────────────────────────────────────┐
│  YOUR MACHINE (running dig +trace)                                   │
│  ──────────────────────────────────                                  │
│  dig performs ITERATIVE queries directly — NOT recursive!            │
│                                                                      │
│  dig → Root → dig → TLD → dig → Authoritative → Answer to you        │
│       ↑_________↑_________↑                                          │
│       These are separate, direct queries from YOUR machine           │
└──────────────────────────────────────────────────────────────────────┘
```